

A Wearable Ergonomic Risk Awareness System for Occupational Settings

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I. INTRODUCTION & MOTIVATION

Occupational musculoskeletal disorders (MSDs), particularly low back pain, are a major global health challenge [1]. According to the Health and Safety Executive, over 500,000 workers suffer work-related back and spine injuries each year, contributing to millions of lost working days [2]. Globally, approximately 1.71 billion people are affected by musculoskeletal conditions. In the European context, WMSDs account for over 40% of all work-related health problems and are a primary cause of sick leave and early retirement among manual workers [3].

Despite advances in wearable sensing technologies, real-world adoption remains low. The issue is not the accuracy of sensors, but how feedback is delivered [4]. In industrial settings, continuous or immediate vibration during physically demanding tasks can distract workers or lead them to ignore the device altogether. As a result, studies indicate that up to 50% of users abandon wearable systems within just two weeks [5]. Practical barriers further limit adoption: workers are often unwilling to wear multiple sensors or use systems requiring frequent calibration, and manual data annotation remains a bottleneck due to high cost, time demands, and privacy concerns [6], [7].

To address these challenges, this project proposes a hybrid system combining a rule-based finite state machine with a lightweight machine learning classifier. Specifically, this project seeks to answer the core research question: **Can a wearable device reliably detect hazardous postures and deliver risk-stratified haptic feedback to avoid alarm fatigue and behavioral rejection?** The system delivers time-delayed haptic nudges rather than continuous vibration, aiming to encourage posture correction without causing distraction or feedback fatigue.

II. RELATED WORK

Inertial measurement units have become the dominant sensing modality for ambulatory human movement monitoring. Mathie et al. [8] established accelerometry as a practical method for long-term continuous monitoring, providing the methodological foundation for subsequent wearable posture systems. Uhlenberg et al. [9] evaluated IMU-based joint angle estimation, reporting median RMSE values of 5–7° against optical motion capture, directly relevant to our trunk flexion threshold design. Ray et al. [10] provided empirical guidance for sensor placement across the body, informing our four-sensor spinal configuration. Münzner et al. [11] showed that CNN-based sensor fusion improves recognition accuracy but at significantly higher computational cost, motivating our choice of a lightweight FSM for on-device classification.

Within the occupational domain, Derungs et al. [13] demonstrated that IMU data combined with regression models can reliably detect technique errors during physical activity, supporting automated ergonomic risk assessment without external infrastructure. Haresamudram et al. [7] identify annotation bottlenecks and cross-subject generalization as persistent open problems, challenges our rule-based approach partially mitigates through calibrated biomechanical thresholds.

Effective posture correction requires feedback users will sustain engaging with over time. Sung et al. [14] demonstrated that vibrotactile feedback should account for body posture during movement to compensate for sensory masking, supporting our two-tier alert design. Rieder et al. [15] found that under compounding negative contexts, users remove devices entirely, motivating a supportive rather than punitive feedback strategy. Künzler et al. [17] showed well-timed interventions increase goal completion by up to 62%, reinforcing the importance of avoiding continuous alerting. Pejović and Musolesi [16] and Torkamaan and Ziegler [19] formalized intelligent prompting and escalating feedback frameworks, providing theoretical

support for our cooldown periods and graduated alert intensity.

III. SYSTEM DESIGN & DATA ACQUISITION

A. System Overview

The wearable ergonomic risk awareness system is built around a three-layer pipeline transforming raw inertial sensor data into real-time haptic feedback. The sensing layer consists of four MPU6050 IMU sensors at the hip, lower back, upper back, and neck. The processing layer runs entirely on the ESP32-S3 microcontroller with no external connectivity. The feedback layer delivers alerts through a vibration motor and LED whenever an unsafe posture is sustained.

B. System Architecture

Fig. 1 shows the physical wiring architecture and Fig. 2 shows the full circuit schematic.

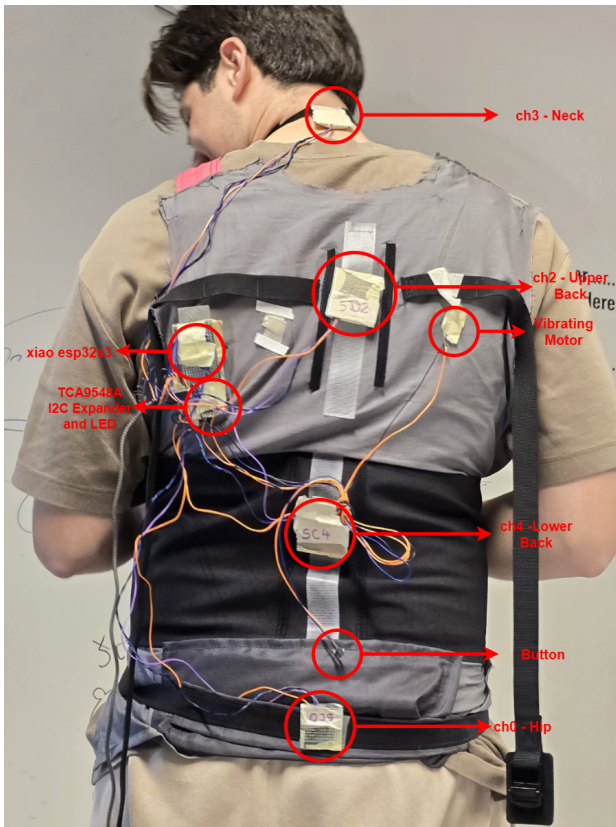


Fig. 1. Hardware wiring architecture showing the ESP32-S3, TCA9548A multiplexer, four MPU6050 IMU sensors on channels CH0-CH3, vibration motor, LED, and push switch.

The **sensing layer** uses four MPU6050 6-axis IMU sensors at hip (CH0), lower back (CH1), upper back (CH2), and neck (CH3). All sensors share the same I2C address (0x68), resolved by a TCA9548A 8-channel multiplexer at address 0x70, fixed by tying address pins A0, A1, A2 to GND. The I2C bus runs at 40 kHz. CH0 provides the pelvic reference for slouch signals Δ_{UB} and Δ_{Neck} ; CH1 contributes to *trunkFlex*; CH2 contributes to both *trunkFlex* and Δ_{UB} ; CH3 provides Δ_{Neck} .

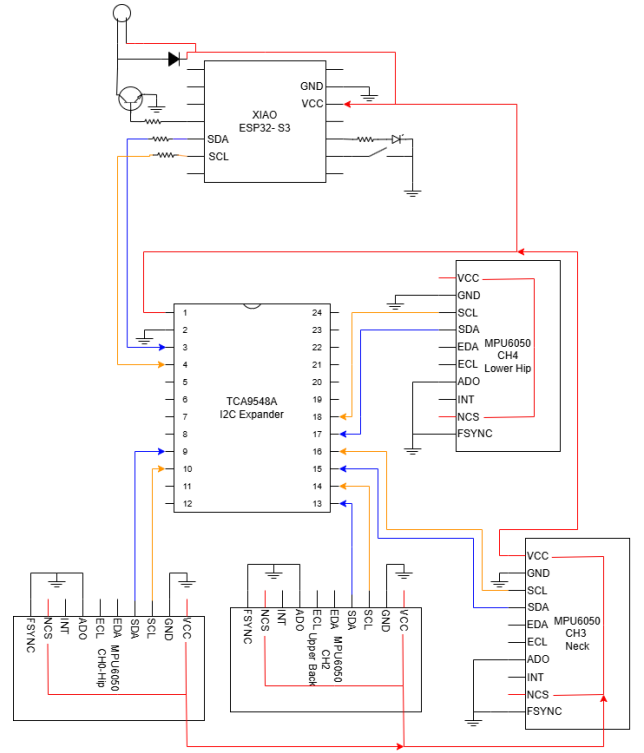


Fig. 2. Circuit schematic showing the NPN transistor motor driver with flyback diode on GPIO 4, the 220 Ω LED current-limiting resistor on GPIO 8, and the internal pull-up switch on GPIO 7.

The **processing layer** is the ESP32-S3 XIAO microcontroller, acting as I2C master, running FSM and ML inference, and driving all output peripherals. The **feedback layer** consists of a vibration motor on GPIO 4 through an NPN transistor (2N2222) with a 1N4007 flyback diode, an LED on GPIO 8 through a 220 Ω resistor, and a push switch on GPIO 7 with internal pull-up.

C. Firmware Flow

On power-up, the system wakes all four MPU6050 sensors and enters a 5-second calibration phase during which the user stands still. Accelerometer samples are averaged per channel to produce a neutral reference angle. The main loop then runs at approximately 100 Hz, reading and validating each sensor by checking vector magnitude within 8000–24000 LSB. Valid readings are converted to pitch angles using atan2 on the normalised X and Z components.

D. Data Acquisition

Data was collected from six group members, each performing neutral standing, forward bending, extreme forward lean, and slouching with upper back rounding. Each activity was held for at least 10 seconds and repeated three times per participant. Ground truth labels were assigned by video observation synchronised with the serial monitor output at approximately 3 Hz. The complete source code is available at <https://github.com/your-repo-link>.

IV. METHODOLOGY

A. Finite State Machine for Posture Classification

Posture detection is implemented as a Finite State Machine (FSM) on the ESP32-S3, classifying posture into *Standing*, *Slouching*, *Bending*, and *Deep Bend*, with a fifth internal state *Prolonged Flexion* triggered when bad posture is sustained beyond a time threshold.

1) *Signal Computation*: Two primary signals drive all state transitions. The first is **trunkFlex**, measuring overall forward trunk inclination:

$$\text{trunkFlex} = \frac{|\text{UB}_{rel}| + |\text{LB}_{rel}|}{2} \quad (1)$$

Absolute values prevent sign cancellation when sensors tilt in opposite directions. The raw value is smoothed:

$$\text{flex}_t = 0.8 \cdot \text{flex}_{t-1} + 0.2 \cdot \text{trunkFlex}_{raw} \quad (2)$$

A 2.0° deadband prevents spurious transitions near neutral. The second signal detects **slouching** using the hip as a reference:

$$\Delta_{UB} = \text{UB}_{rel} - \text{Hip}_{rel}, \quad \Delta_{Neck} = \text{Neck}_{rel} - \text{Hip}_{rel} \quad (3)$$

$$\text{isSlouch} = (\text{trunkFlex} < 25^\circ) \wedge (|\Delta_{UB}| > 8^\circ) \wedge (|\Delta_{Neck}| > 8^\circ) \quad (4)$$

A sanity check rejects readings above 180° to prevent hardware spikes corrupting the FSM.

2) *State Definitions and Transitions*: **Standing**: all sensors near zero; return condition is $\text{trunkFlex} < 10^\circ$ held for 600 ms. **Slouching**: entered when *isSlouch* is true; upper back rounds while hip stays upright. **Bending**: $\text{trunkFlex} > 25^\circ$; safe short term, no feedback. **Deep Bend**: $\text{trunkFlex} > 70^\circ$; dangerous compressive load, FSM only. **Prolonged Flexion**: any bad posture held > 5000 ms; triggers feedback. Thresholds are summarised in Table I and transitions in Fig. 3.

TABLE I
FSM STATE TRANSITION THRESHOLDS

Condition	Threshold
Standing (return)	$\text{trunkFlex} < 10^\circ$
Bending onset	$\text{trunkFlex} > 25^\circ$
Deep Bend onset	$\text{trunkFlex} > 70^\circ$
Slouch (UB)	$ \Delta_{UB} > 8^\circ$
Slouch (Neck)	$ \Delta_{Neck} > 8^\circ$
Feedback delay	5000 ms
Stabilisation	600 ms

3) *Feedback Mechanism*: When bad posture is sustained for 5000 ms, the motor activates for a duration encoding severity: Slouching triggers a 300 ms pulse; Deep Bend and Prolonged Flexion trigger an 800 ms pulse. A 3000 ms cooldown prevents alarm fatigue [17], [19]. Motor and LED are always driven together.

4) *Calibration*: The user presses a button to initiate a 5-second calibration phase standing upright and still. Reference angles are averaged per channel and subtracted from all subsequent readings, allowing the system to adapt to different body shapes without manual threshold adjustment.

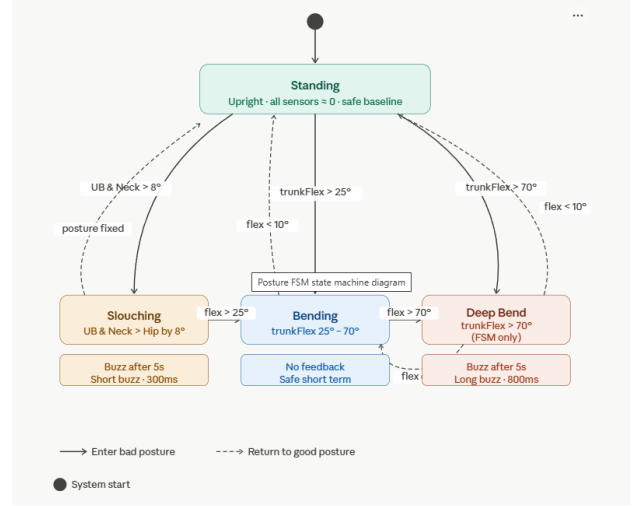


Fig. 3. FSM state transition diagram for posture classification.

B. Machine Learning Classifier

1) *Data Collection and Acquisition*: Sensor data was streamed to Edge Impulse at 45 Hz via serial communication. Data was recorded in 5-second windows covering three posture classes: standing, bending, and slouching. Approximately 15 minutes of labelled data were collected from all team members, with natural movement variations introduced to improve generalisation.

2) *Pre-processing and Feature Extraction*: Raw IMU signals were pre-processed using a 32-point FFT to convert time-domain signals to the frequency domain (Fig. 4, Fig. 5). Spectral features were chosen because human motion patterns exhibit distinct frequency components, reducing noise sensitivity and improving class separability while remaining computationally efficient for embedded deployment.

Spectral power (log)

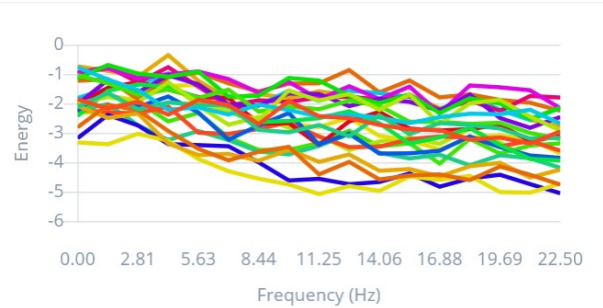


Fig. 4. Spectral power of IMU signals across posture classes.

3) *Model Architecture and Training*: A feedforward neural network (FNN) was trained on the extracted features with an input layer of 504 features, four fully connected hidden layers, and an output layer of 3 nodes (standing, bending, slouching). The model was trained for 60 epochs. A FNN was selected over CNNs or RNNs due to its balance between classification

Feature explorer

- bending
- slouching
- standing

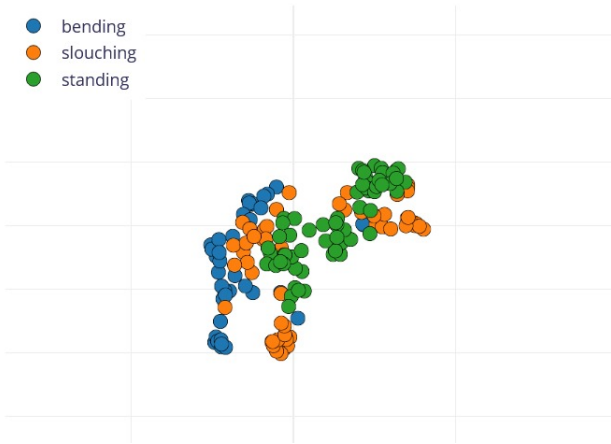


Fig. 5. Feature map showing class separability in the frequency domain.

capability and computational efficiency on the ESP32. Fig. 6 shows the architecture and Fig. 7 shows the training curves.

Neural network architecture

Neural network Complete architecture / classic ML



Fig. 6. FNN model architecture with 504-feature input and 3-class output.

4) *Model Optimization and Deployment:* The model was quantized to INT8 precision before deployment, reducing model size and enabling real-time inference on the ESP32. INT8 quantization was chosen to ensure compatibility with embedded hardware memory and processing constraints. The final quantized model achieved a validation accuracy of 74.2%.

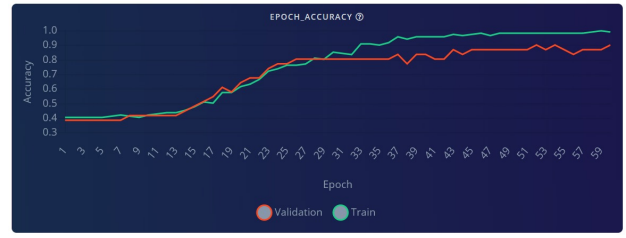


Fig. 7. Training and validation accuracy curves (Float32).

V. RESULTS & DISCUSSION

A. FSM Performance

The FSM was evaluated through real-time testing across all four posture states using serial monitor logs to verify state transitions and timing behaviour.

Fig. 8 shows a session beginning with calibration completing with 192 valid samples. Following calibration, the system entered STANDING with all sensor values near zero, confirming correct reference subtraction. HipVsUB and HipVsNeck remained below 2.1° and 2.0° respectively, well below the 8° slouch threshold.

Fig. 9 shows DEEP BEND with trunkFlex consistently between 72° and 84° , well above the 70° threshold. The 5-second countdown progresses from “buzz in 4s” to “buzz in 0s”, confirming the Prolonged Flexion timing mechanism. The motor activated correctly at each countdown end.

```
Press button to start calibration...
Stand STRAIGHT and stay still for 5 seconds...
Calibrating... 5s remaining
Calibrating... 4s remaining
Calibrating... 3s remaining
Calibrating... 2s remaining
Calibrating... 1s remaining
Calibration done (192 good samples). Monitoring started.

Hip:0.4 LB:-0.8 UB:-0.4 Neck:-0.6 TF:0.1 raw:0.6 HipVsUB:0.8 HipVsNeck:1.0 State: STANDING
Hip:0.3 LB:-1.1 UB:-0.6 Neck:-0.4 TF:0.7 raw:0.9 HipVsUB:0.9 HipVsNeck:0.7 State: STANDING
Hip:0.1 LB:-0.9 UB:-0.8 Neck:-1.0 TF:0.8 raw:0.9 HipVsUB:0.9 HipVsNeck:1.1 State: STANDING
Hip:0.3 LB:-1.1 UB:-0.5 Neck:-0.7 TF:0.7 raw:0.8 HipVsUB:0.8 HipVsNeck:1.0 State: STANDING
Hip:-0.1 LB:-0.4 UB:-0.7 Neck:-1.2 TF:0.6 raw:0.6 HipVsUB:0.7 HipVsNeck:1.2 State: STANDING
Hip:0.3 LB:-0.4 UB:-0.2 Neck:-0.2 TF:0.4 raw:0.3 HipVsUB:0.6 HipVsNeck:0.5 State: STANDING
Hip:-0.2 LB:0.0 UB:0.0 Neck:-1.3 TF:0.3 raw:0.0 HipVsUB:0.2 HipVsNeck:1.0 State: STANDING
Hip:-0.3 LB:-0.3 UB:-0.2 Neck:-1.0 TF:0.3 raw:0.2 HipVsUB:0.0 HipVsNeck:0.8 State: STANDING
Hip:-0.5 LB:-0.3 UB:0.0 Neck:-0.2 TF:0.2 raw:0.1 HipVsUB:0.6 HipVsNeck:0.3 State: STANDING
Hip:-1.0 LB:-0.9 UB:-0.6 Neck:-0.8 TF:0.7 raw:0.8 HipVsUB:0.4 HipVsNeck:0.2 State: STANDING
Hip:-0.6 LB:1.3 UB:0.2 Neck:0.2 TF:0.6 raw:0.7 HipVsUB:0.8 HipVsNeck:0.8 State: STANDING
Hip:2.0 LB:0.1 UB:2.2 Neck:1.4 TF:0.9 raw:1.2 HipVsUB:0.3 HipVsNeck:0.6 State: STANDING
Hip:0.2 LB:-0.0 UB:0.3 Neck:-0.1 TF:0.5 raw:0.2 HipVsUB:0.1 HipVsNeck:0.2 State: STANDING
Hip:0.1 LB:0.6 UB:0.1 Neck:-0.1 TF:0.4 raw:0.4 HipVsUB:0.0 HipVsNeck:0.1 State: STANDING
Hip:-0.7 LB:1.3 UB:-0.4 Neck:-0.6 TF:0.5 raw:0.9 HipVsUB:0.3 HipVsNeck:0.1 State: STANDING
Hip:-1.6 LB:-1.5 UB:0.6 Neck:-0.1 TF:1.8 raw:1.0 HipVsUB:2.1 HipVsNeck:1.5 State: STANDING
Hip:0.1 LB:0.9 UB:0.9 Neck:-0.5 TF:0.8 raw:0.9 HipVsUB:0.8 HipVsNeck:0.6 State: STANDING
Hip:0.7 LB:1.5 UB:-0.2 Neck:-1.3 TF:0.9 raw:0.9 HipVsUB:0.4 HipVsNeck:2.0 State: STANDING
```

Fig. 8. Serial monitor output showing calibration with 192 good samples and stable STANDING classification.

```
Posture changed to: DEEP BEND
Hip:15.8 LB:-82.3 UB:-82.9 Neck:-55.6 TF:79.7 raw:82.4 HipVsUB:102.7 HipVsNeck:115.4 State: DEEP BEND! [0s in posture, buzz in 4s]
Hip:21.7 LB:-82.5 UB:-82.4 Neck:-56.1 TF:83.7 raw:82.4 HipVsUB:104.0 HipVsNeck:117.7 State: DEEP BEND! [0s in posture, buzz in 4s]
Hip:19.6 LB:-79.9 UB:-78.6 Neck:-52.4 TF:79.3 raw:78.8 HipVsUB:98.3 HipVsNeck:112.0 State: DEEP BEND! [0s in posture, buzz in 4s]
Hip:16.7 LB:-76.0 UB:-76.5 Neck:-49.6 TF:75.5 raw:76.3 HipVsUB:93.2 HipVsNeck:105.6 State: DEEP BEND! [1s in posture, buzz in 3s]
Hip:16.3 LB:-72.5 UB:-73.4 Neck:-46.7 TF:74.5 raw:72.9 HipVsUB:89.7 HipVsNeck:103.1 State: DEEP BEND! [1s in posture, buzz in 3s]
Hip:16.4 LB:-71.8 UB:-72.6 Neck:-45.9 TF:72.0 raw:72.2 HipVsUB:89.0 HipVsNeck:102.3 State: DEEP BEND! [1s in posture, buzz in 3s]
Hip:16.5 LB:-74.3 UB:-74.0 Neck:-46.7 TF:73.3 raw:73.5 HipVsUB:90.5 HipVsNeck:103.2 State: DEEP BEND! [2s in posture, buzz in 2s]
Hip:16.1 LB:-72.9 UB:-73.6 Neck:-46.5 TF:73.2 raw:73.2 HipVsUB:89.7 HipVsNeck:102.6 State: DEEP BEND! [2s in posture, buzz in 2s]
Hip:16.1 LB:-72.9 UB:-73.7 Neck:-46.1 TF:73.2 raw:73.3 HipVsUB:89.8 HipVsNeck:102.2 State: DEEP BEND! [2s in posture, buzz in 2s]
Hip:17.3 LB:-73.9 UB:-75.4 Neck:-47.6 TF:74.5 raw:74.3 HipVsUB:92.1 HipVsNeck:104.9 State: DEEP BEND! [3s in posture, buzz in 1s]
Hip:17.5 LB:-73.8 UB:-74.7 Neck:-47.7 TF:74.5 raw:74.3 HipVsUB:92.1 HipVsNeck:105.1 State: DEEP BEND! [3s in posture, buzz in 1s]
Hip:17.7 LB:-74.7 UB:-75.5 Neck:-49.2 TF:74.9 raw:75.1 HipVsUB:93.2 HipVsNeck:106.9 State: DEEP BEND! [3s in posture, buzz in 1s]
Hip:17.3 LB:-73.9 UB:-75.4 Neck:-47.6 TF:74.5 raw:74.3 HipVsUB:92.1 HipVsNeck:104.9 State: DEEP BEND! [4s in posture, buzz in 0s]
Hip:17.3 LB:-74.6 UB:-75.4 Neck:-48.6 TF:74.8 raw:75.0 HipVsUB:92.7 HipVsNeck:105.9 State: DEEP BEND! [4s in posture, buzz in 0s]
Hip:18.0 LB:-75.0 UB:-75.4 Neck:-48.6 TF:75.0 raw:75.2 HipVsUB:93.4 HipVsNeck:106.6 State: DEEP BEND! [4s in posture, buzz in 0s]
```

Fig. 9. Serial monitor output showing DEEP BEND with trunkFlex 72° – 84° and 5-second feedback countdown.

Bending was consistently detected for trunkFlex between 25° and 70° with no haptic feedback triggered. Slouching was

reliably identified using relative angle differences, ensuring clear separation from uniform forward bending. Response delay was approximately 200–500 ms due to the low-pass filter and stabilisation period, which eliminates rapid oscillations at the cost of slight lag. Across all sessions, FSM transitions matched expected posture conditions without misclassification in observed scenarios. These results demonstrate that reliable on-device posture monitoring is feasible without cloud connectivity, directly addressing the infrastructure-free deployment requirements of pervasive computing in industrial settings.

B. Machine Learning Classifier Performance

The final INT8 model achieved a validation accuracy of 74.2%. The confusion matrix (Fig. 10) and metrics table (Fig. 11) show strong performance distinguishing standing from dynamic postures, with some confusion between bending and slouching due to similar trunk orientations. The model performed inference on the ESP32 within milliseconds, confirming suitability for near real-time feedback. The weighted average F1-score was 0.74, with precision and recall both at 0.74.

Primary sources of error were: (1) spectral feature overlap between bending and slouching due to similar forward inclination; (2) limited dataset size (15 minutes) restricting cross-user generalisation; (3) INT8 quantization reducing numerical precision; and (4) loose sensor mounting introducing orientation noise. Despite these limitations, the results demonstrate practical viability of combining lightweight ML with embedded hardware for real-time posture monitoring.

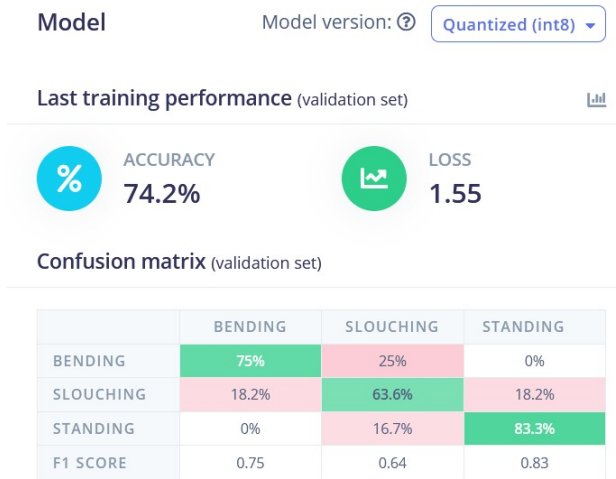


Fig. 10. INT8 model confusion matrix and per-class accuracy.

C. Limitations

System performance depends on accurate sensor placement and calibration per user. The smoothing filter prioritises stability over responsiveness, meaning rapid movements may temporarily cause inaccuracies. Future work should include a formal user study with labelled ground truth to produce full

Metrics (validation set)

METRIC	VALUE
Area under ROC Curve ?	0.91
Weighted average Precision ?	0.74
Weighted average Recall ?	0.74
Weighted average F1 score ?	0.74

Fig. 11. Validation set metrics: precision, recall, F1-score, and AUC.

per-class accuracy metrics across a larger subject pool, and a purpose-built vest with integrated sensor mounting to eliminate noise from sensor movement.

VI. CONCLUSION

This work presents a wearable ergonomic monitoring system integrating four MPU6050 IMU sensors with an ESP32-S3 microcontroller, supporting both rule-based FSM and feed-forward neural network posture detection entirely on-device. Key contributions include biomechanical feature extraction, temporal reasoning through a five-state FSM, and a human-centered feedback strategy following JITAI principles [17], [19] that minimizes alarm fatigue by delivering interventions only after sustained unsafe postures. The evaluation shows hardware integration is the primary performance determinant: sensor placement instability introduced orientation noise limiting both FSM and ML reliability. The ML model achieved 74.2% validation accuracy, with main confusion between bending and slouching. Future work should address vest-integrated sensor mounting, expanded data collection, and real industrial deployment studies.

VII. AUTHOR CONTRIBUTIONS

Mansi Bhardwaj: Hardware design, FSM implementation, data collection.

Afeef Ahmed: Machine learning, data collection.

Sina Shariati: FSM implementation, machine learning, data collection.

Ming-Hua Hsieh: Vest design and production, data collection.

Hari Sahar Ravi Kumar: Hardware design, data collection.

Andrea Battarelli Martini: Vest design and production, data collection.

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